

Impact of Underground Cables on Harmonic Hosting Capacity

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Abstract—In the context of increasing power electronic interfaced assets and underground cable connections, this paper presents a novel optimal harmonic hosting capacity method for assignment of individual harmonic current limits to units in a phase-balanced three-phase power system. The approach is compared to those in the relevant standards: IEEE519-2022 and IEC61000-3-6:2008. The numerical results show that the proposed approach better utilizes the power system compared to the standards. Furthermore, the proposed approach avoids harmonic voltage violations in the presence of resonances, in contrast to the standards. This is especially important in power systems with increasing share of underground cable connections.

Index Terms—Harmonic distortion, underground cables, hosting capacity, mathematical optimization.

I. INTRODUCTION

A. Background, Motivation and Literature Review

Reaching a net zero economy requires a massive undertaking in electrification of the energy system. The majority of the effort occurs on three fronts:

- 1) shift towards renewable power generation, e.g., wind and photovoltaic generation,
- 2) electrification of energy demand, e.g., transport, heating and industrial processes, and
- 3) expansion of the transmission system to facilitate the increase in electric load and generation, and the geographic shift of latter.

In the context of power quality this brings about two major shifts. First, renewable energy sources and new types of electrical loads are almost exclusively power electronic interfaced assets. Power electronic interfaced assets behave fundamentally different compared to classical electrical machines, or resistive and inductive demand. One of the major concerns is the increase in injected current at non-fundamental frequencies, i.e., harmonics. Second, due to societal pressure, in high density areas, underground cables have become the preferred option for transmission system expansion in favor of overhead lines. As a consequence, the impedance of the transmission system shifts from a purely inductive impedance, i.e., linearly increasing with frequency, to an impedance containing resonances, i.e., going to zero or infinity at certain

frequencies. Sun et al. highlight that the traditional harmonic studies performed at the initial planning stage may no longer be appropriate in this context, [1]. Considering the aforementioned transitions, this paper aims at answering the question: “what is the maximum harmonic distortion, e.g., number of power electronic converters, that can be reliably hosted in a given power system, with increasing degrees of underground cable connections?”. In literature, this concept is referred to as the harmonic hosting capacity of a power system [2], [3].

There are several standards and, more recently, limited academic literature presenting methods to determine the harmonic hosting capacity of a power system, and differing in a number of features, including network representation. The methods in the relevant standards, e.g., IEEE519-2022 [4] and IEC61000-3-6:2008 [5], use a nodal representation of the network, i.e., calculate the harmonic hosting capacity for each individual bus, (partially) omitting the interaction between different busses. The standard IEC61000-3-6:2008, at pg. 35, partially compensates for omitting this interaction by the introduction of influence factors, adding significant complexity. Furthermore, in the presence of series resonances, i.e., where the nodal impedance goes to zero, the method described in the standard is unable to ensure compliance with the prescribed individual harmonic voltage limits. The standard IEC61000-3-6:2008, at pg. 23, partially compensates the series resonance effects by disregarding the actual harmonic impedance for around the series resonance frequency, and replacing it with the fundamental impedance scaled with the harmonic number.

Recent literature shifted towards a full network representation in the context of harmonic hosting capacity. Sun et al. present a non-linear optimization method, including harmonic voltage constraints, to determine the hosting capacity for distributed generation in distribution networks [6], [7]. Sakar et al. present a monte-carlo power flow method to determine the harmonic hosting capacity for large-scale photovoltaic plants [8] whereas Carmelito presents a similar approach to determine the harmonic hosting capacity for electric vehicles [9]. In contrast to Sakar et al., Carmelito considers an unbalanced power flow rather than a balanced one.

B. Contribution and Outline

In this paper, a holistic one-shot optimization method is presented to exactly and tractably determine the lower

bound of the steady-state harmonic hosting capacity for a phase-balanced three-phase power system. The problem is formulated from the perspective of the power system operator, i.e., without knowledge of the (future) assets of its costumers. Similar to the methods described in the relevant standards, the harmonic hosting capacity is determined in terms of harmonic current magnitude. The models and case study presented in the paper are made available as part of the JULIA-package: HARMONICPOWERMODELS.JL, at: <https://github.com/timmyfaraday/HarmonicPowerModels.jl>.

The remainder of the paper is organized as follows: Section II presents the mathematical framework of the methods in the relevant standards and the novel optimization method. Section III shows results for the IEEE-30 bus system. Section IV concludes the paper.

II. MATHEMATICAL FRAMEWORK

The harmonic hosting capacity problem determines the maximum amount of harmonic current that can be injected by units $u \in \mathcal{U}$, an umbrella term used for load and generation, without violating any voltage and current limits. A harmonic voltage or current oscillates at a multiple $h \in \mathcal{H} \subset \mathbb{N}$ of the fundamental frequency, e.g., 50 Hz or 60 Hz. This section presents three different methods to determine the harmonic hosting capacity in terms of individual harmonic current magnitude $I_{u,h}^{\text{ihd}}$ that can be injected by every unit of a given phase-balanced three-phase power system:

- § II-A look-up method to determine the *nodal* harmonic hosting capacity as per std. IEEE519-2022,
- § II-B analytical method to calculate the *nodal* harmonic hosting capacity as per std. IEC61000-3-6:2008, and
- § II-C novel *global* one-shot optimization method to calculate the harmonic hosting capacity.

It is assumed that no harmonic recombination occurs for all three methods providing the worst-case harmonic hosting capacities. The following notation is used throughout the mathematical framework.

element	notation	example
index	lower-case italic	x
set	calligraphic	$\mathcal{X}$
variable	sans serif	X
parameter	italic	$\bar{X}$
complex quant.	bar notation	$\bar{X}, \bar{X}$

A. Look-Up Method Following Std. IEEE519-2022

The IEEE519-2022 standard provides individual harmonic current limits $I_{u,h}^{\text{ihd}}$ for any unit u with a) a mix of both linear and non-linear loads, and b) rated generation less than 10 % of annual average load, relative to the fundamental unit current magnitude, depending on: a) rated voltage magnitude, and b) ratio between short-circuit and maximum load current, at the point-of-common-coupling. For the actual values, in general, the reader is referred to Tables 2-4 of the std. IEEE519-2022, respectively.

B. Analytical Method Following Std. IEC61000-3-6:2008

The standard IEC61000-3-6:2008 describes a three stage approach for limiting individual harmonic current injection thus ensuring the voltage distortion to remain below the planning levels. We focus on stage two for (extra) high-voltage power systems, evaluating the emission limits relative to the actual system characteristics. Before allocating emission limits to individual units, the common planning harmonic voltage limits are shared between the unit buses $i \in \mathcal{I}^u \subset \mathcal{I}$. The standard proposes to distribute these planning levels based on the (future) connected nodal apparent power S_i^{tot} and influence coefficients $K_{ij,h}$,

$$U_{i,h}^{\text{ihd}} \leq \frac{S_i^{\text{tot}}}{\sum_{j \in \mathcal{I}^u} K_{ij,h} S_j^{\text{tot}}} U_h^{\text{ihd}}, \quad \forall i \in \mathcal{I}^u, h \in \mathcal{H}. \quad (1)$$

The individual harmonic voltage limits of the unit buses are further divided among the units u connected to that unit bus i ,

$$U_{u,h}^{\text{ihd}} = U_{i,h}^{\text{ihd}} \frac{S_u}{S_u^{\text{tot}}}, \quad \forall ui \in \mathcal{T}^u, h \in \mathcal{H}, \quad (2)$$

and, the harmonic current limits $I_{u,h}^{\text{ihd}}$ for the individual unit u is determined using the harmonic impedance $Z_{i,h}$ at bus i :

$$I_{u,h}^{\text{ihd}} = \frac{U_{u,h}^{\text{ihd}}}{Z_{i,h}}, \quad \forall ui \in \mathcal{T}^u, h \in \mathcal{H}. \quad (3)$$

C. Novel Global One-Shot Optimization Method

The global one-shot optimal power flow framework for determining the harmonic hosting capacity exploits the fact that harmonic power flows can be decoupled as multiple independent power flows for each harmonic $h \in \mathcal{H}$ (proof: [10] § 16.6). The harmonic power flows can best be represented using the rectangular current-voltage formulation [11] and the power system is represented by an extended graph $(\mathcal{I}, \mathcal{E}, \mathcal{U})_h$ for each harmonic $h \in \mathcal{H}$, using following variables.

name	base-set	tuple-set	variable
bus voltage	$i \in \mathcal{I}$	-	$\bar{U}_{i,h} = U_{i,h}^{\text{re}} + j U_{i,h}^{\text{im}}$
edge current	$e \in \mathcal{E}$	$ej \in \mathcal{T}^e$	$\bar{I}_{ej,h} = (I_{ej,h}^{\text{sh, re}} + I_{ej,h}^{\text{s, re}}) + j (I_{ej,h}^{\text{sh, im}} + I_{ej,h}^{\text{s, im}})$, this implies $\bar{I}_{ej,h}^{\text{s}} + \bar{I}_{ej,h}^{\text{sh}} = 0$
unit current	$u \in \mathcal{U}$	$ui \in \mathcal{T}^u$	$\bar{I}_{u,h} = I_{u,h}^{\text{re}} + j I_{u,h}^{\text{im}}$

In the rectangular current-voltage formulation, the set of linear equality constraints (4)-(20) uniquely describing the real-valued feasible set for the harmonic power flow in the power system is,

BUS CONSTRAINTS

reference bus voltage – $\forall i \in \mathcal{I}^{\text{ref}}, h \in \mathcal{H}$:

$$U_{i,h}^{\text{re}} = U_{i,h}^{\text{ref, re}}, \quad (4)$$

$$U_{i,h}^{\text{im}} = U_{i,h}^{\text{ref, im}}, \quad (5)$$

Kirchhoff's current law – $\forall i \in \mathcal{I}, h \in \mathcal{H}$:

$$\sum_{eij \in \mathcal{T}^e} l_{eij,h}^{\text{re}} + \sum_{ui \in \mathcal{T}^u} l_{ui,h}^{\text{re}} + g_{i,h}^{\text{sh}} U_{i,h}^{\text{re}} - b_{i,h}^{\text{sh}} U_{i,h}^{\text{im}} = 0, \quad (6)$$

$$\sum_{eij \in \mathcal{T}^e} l_{eij,h}^{\text{im}} + \sum_{ui \in \mathcal{T}^u} l_{ui,h}^{\text{im}} + g_{i,h}^{\text{sh}} U_{i,h}^{\text{im}} + b_{i,h}^{\text{sh}} U_{i,h}^{\text{re}} = 0, \quad (7)$$

root mean square voltage – $\forall i \in \mathcal{I}$:

$$(U^{\min})^2 \leq \sum_{h \in \mathcal{H}} (U_{i,h}^{\text{re}})^2 + (U_{i,h}^{\text{im}})^2 \leq (U^{\max})^2, \quad (8)$$

total harmonic voltage distortion – $\forall i \in \mathcal{I}$:

$$\sum_{h \in \mathcal{H} \setminus \{1\}} (U_{i,h}^{\text{re}})^2 + (U_{i,h}^{\text{im}})^2 \leq (U^{\text{thd}})^2 ((U_{i,1}^{\text{re}})^2 + (U_{i,1}^{\text{im}})^2), \quad (9)$$

individual harmonic voltage distortion – $\forall i \in \mathcal{I}, h \in \mathcal{H} \setminus \{1\}$:

$$(U_{i,h}^{\text{re}})^2 + (U_{i,h}^{\text{im}})^2 \leq (U_h^{\text{ihd}})^2 ((U_{i,1}^{\text{re}})^2 + (U_{i,1}^{\text{im}})^2), \quad (10)$$

where $U^{\min}$, $U^{\max}$, U^{thd} , and U_h^{ihd} denote the minimum and maximum root-mean-square voltage magnitude, total harmonic voltage distortion limit, and individual harmonic voltage distortion limit, respectively.

EDGE CONSTRAINTS

Ohm's law – $\forall eij \in \mathcal{T}^{e\rightarrow}, h \in \mathcal{H}$:

$$U_{j,h}^{\text{re}} = U_{i,h}^{\text{re}} - r_{e,h}^{\text{s}} l_{eij,h}^{\text{s, re}} + x_{e,h}^{\text{s}} l_{eij,h}^{\text{s, im}}, \quad (11)$$

$$U_{j,h}^{\text{im}} = U_{i,h}^{\text{im}} - r_{e,h}^{\text{s}} l_{eij,h}^{\text{s, im}} - x_{e,h}^{\text{s}} l_{eij,h}^{\text{s, re}}, \quad (12)$$

branch total current – $\forall eij \in \mathcal{T}^e, h \in \mathcal{H}$:

$$l_{eij,h}^{\text{re}} = g_{eij,h}^{\text{sh}} U_{i,h}^{\text{re}} - b_{eij,h}^{\text{sh}} U_{i,h}^{\text{im}} + l_{eij,h}^{\text{s, re}}, \quad (13)$$

$$l_{eij,h}^{\text{im}} = g_{eij,h}^{\text{sh}} U_{i,h}^{\text{im}} + b_{eij,h}^{\text{sh}} U_{i,h}^{\text{re}} + l_{eij,h}^{\text{s, im}}, \quad (14)$$

root mean square current – $\forall eij \in \mathcal{T}^e$:

$$\sum_{h \in \mathcal{H}} (l_{eij,h}^{\text{re}})^2 + (l_{eij,h}^{\text{im}})^2 \leq (I_e^{\text{rated}})^2, \quad (15)$$

where $r_{e,h}^{\text{s}}$, $x_{e,h}^{\text{s}}$, $g_{e,h}^{\text{sh}}$ and $b_{e,h}^{\text{sh}}$ denote the series resistance, series reactance, shunt conductance, and shunt susceptance for a given harmonic $h \in \mathcal{H}$, respectively. Additionally, I_e^{rated} denotes the rated current limit of an edge $e \in \mathcal{E}$.

UNIT CONSTRAINTS

fundamental power limits – $\forall ui \in \mathcal{T}^u$:

$$P_u^{\min} \leq U_{i,1}^{\text{re}} l_{u,1}^{\text{re}} + U_{i,1}^{\text{im}} l_{u,1}^{\text{im}} \leq P_u^{\max}, \quad (16)$$

$$Q_u^{\min} \leq U_{i,1}^{\text{im}} l_{u,1}^{\text{re}} - U_{i,1}^{\text{re}} l_{u,1}^{\text{im}} \leq Q_u^{\max}, \quad (17)$$

harmonic current injection – $\forall u \in \mathcal{U}^h, h \in \mathcal{H} \setminus \{1\}$:

$$\cos(\phi_{u,h}^{\text{ref}}) l_{u,h}^{\text{ihd}} = l_{u,h}^{\text{re}}, \quad (18)$$

$$\sin(\phi_{u,h}^{\text{ref}}) l_{u,h}^{\text{ihd}} = l_{u,h}^{\text{im}}. \quad (19)$$

absolute equality criterion – $\forall u, v \in \mathcal{U}^h, h \in \mathcal{H} \setminus \{1\}$:

$$l_{u,h}^{\text{ihd}} = l_{v,h}^{\text{ihd}}, \quad (20)$$

where $P_u^{\min}$, $P_u^{\max}$, $Q_u^{\min}$, and $Q_u^{\max}$ denote the minimum and maximum, active and reactive power of a unit $u \in \mathcal{U}$, respectively. This comprises fixed power units by setting the minimum and maximum equal. Additionally, $\phi_{u,h}^{\text{ref}}$ denotes the reference angle for harmonic current injection. The worst-case harmonic hosting capacity occurs when all harmonic current

injection $\bar{l}_{u,h}$ are in phase. The absolute equality constraint in (20) establishes a lower bound on the harmonic hosting capacity as each unit gets the harmonic budget, which is limited by the most constrained unit.

Finally, the optimal harmonic hosting capacity in terms of individual harmonic current injections $l_{u,h}^{\text{ihd}}$ is determined as,

$$\max \sum_{u \in \mathcal{U}, h \in \mathcal{H} \setminus \{1\}} l_{u,h}^{\text{ihd}}, \quad \text{s.t. (4) – (20)}. \quad (21)$$

III. CASE STUDY

This section presents a case study comparing three methods to determine the harmonic hosting capacity of a phase-balanced three-phase power system, using the IEEE 30 bus test system [12]. First, the necessary technical assumptions for the methods are enumerated. Second, the harmonic hosting capacity is determined for a purely inductive network based on the three methods. Lastly, the impact of underground cable connections on the harmonic hosting capacity is investigated. Due to space limitations, it is impossible to show all resulting individual harmonic unit currents and bus voltages for all three methods, consequently, specific results are selected to support the conclusions. However, the case study has been made available in HARMONICPOWERMODELS.JL

A. Technical Assumptions

For consistency the individual harmonic voltage limits U_h^{ihd} denoted in IEEE519:2022 are used for all three methods.

1) *Look-Up Method Following Std. IEEE519-2022*: To look-up the relative individual harmonic current limits $l_{u,h}^{\text{ihd}}$, the nodal three-phase short-circuit current is required. They are calculated under following assumptions: a) no short-circuit contribution of the load units, and b) generator short circuit power is chosen to be 2.0GVA for the first two generators and 0.2GVA for all others, all with a XR-ratio of 20.

2) *Analytical Method Following Std. IEC61000-3-6:2008*: To most-clearly illustrate the difference between a nodal and a full network approach to determine the harmonic hosting capacity, all harmonic influence factors $K_{ij,h}$ are set to one. Furthermore, the reader is reminded that no harmonic recombination is assumed, consequently, all parameters α related to the general summation law are set to one.

3) *Novel Optimization Method*: To be consistent with all other methods, root-mean-square or total harmonic distortion limits are not enforced. Furthermore, the absolute equality is enforced between all units, consequently, all individual harmonic current injection for all units will be equal.

B. Case A: Inductive Network

This section presents the results for a purely inductive network for all three methods, in terms of individual harmonic current injection $l_{u,h}^{\text{ihd}}$ by the units and the resulting individual harmonic bus voltage $U_{i,h}^{\text{ihd}}$. All shunt susceptances of the branches in the IEEE 30 bus system are set to zero to obtain a purely inductive network impedance, i.e., exhibiting no resonances and increasing linearly with the frequency. The

nodal harmonic impedance of high-voltage bus 4 and medium-voltage bus 17 are illustrated in Fig. 1.

Fig. 2 shows the individual harmonic current injection $I_{u,h}^{hd}$ by the units for all three methods and Fig. 3 shows that the presented optimization approach best utilizes the prescribed individual harmonic voltage limits. The IEEE519-2022 method results in harmonic voltages exceeding the prescribed limits at certain harmonics for the medium-voltage bus 17 whereas, the IEC61000-3-6:2008 method results in harmonic bus voltages significantly lower than the prescribed limits. Consequently, in the context of a purely inductive network, the presented optimization method best utilizes the harmonic hosting capacity of the network without violating any individual harmonic voltage limit.

C. Case B: Underground Cable Network

This section presents the results for a network with underground cable connections for all three methods, in terms of individual harmonic current injection $I_{u,h}^{hd}$ by the units and the resulting individual harmonic bus voltage $U_{i,h}^{hd}$. All shunt susceptances of the edges are restored to the original data in the IEEE 30 bus system. This adds a number of resonances to the nodal harmonic impedance, as shown in Fig. 1. Given that all underground cable connections are at the high-voltage level, these resonances are significantly more pronounced for the high-voltage bus 4 compared to the medium-voltage bus 17.

Fig. 4 shows the individual harmonic current injection $I_{u,h}^{hd}$ by the units for all three methods. It can be seen that all methods except for the IEEE519-2022 accounts for the harmonic impedance resonances in the system. Similar as for the purely inductive network, Fig. 5 shows that the presented optimization approach best utilizes the prescribed individual harmonic voltage limits. The IEEE519-2022 method results in harmonic voltages exceeding the prescribed limits at certain harmonics for both the medium-voltage bus 17 and high-voltage. Around the resonance frequency, the IEC61000-3-6:2008 method also results in harmonic voltages exceeding the prescribed limits whereas elsewhere the harmonic bus

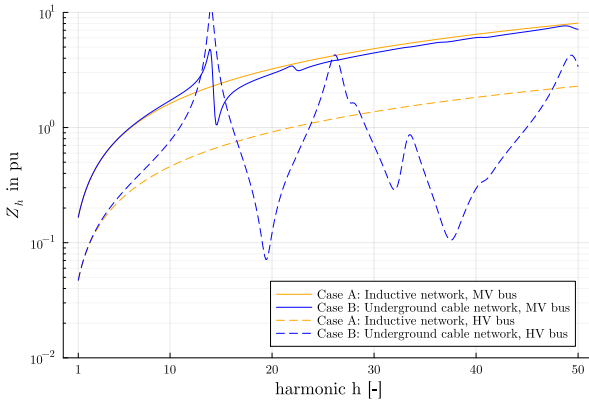


Fig. 1. Nodal impedance of medium-voltage bus 17 (solid lines) and high-voltage bus 4 (dashed lines), considering a) a purely inductive network (blue line), and b) a network with underground cable connections (orange line).

voltages remain significantly lower than the prescribed limits. Consequently, in the context of a network exhibiting resonances, the presented optimization method again best utilizes the harmonic hosting capacity of the network without violating any individual harmonic voltage limit.

IV. CONCLUSION

This paper presents a holistic one-shot optimization method to determine the harmonic hosting capacity of a power system. The presented approach better utilizes the power system compared to the standards IEEE519-2022 and IEC61000-3-6:2008 which tend to over- or underestimate the harmonic hosting capacity, respectively. More importantly the presented method does not lead to any harmonic voltage limit violations, which the existing standards miss to capture near resonances. In power systems with increasing share of underground cable connections, the shift in the resonance frequency amplifies this effect. In this work, the obtained harmonic hosting capacity equally distributes the available harmonic budget among all units whereas future work will investigate different fairness criteria for better utilizing the harmonic hosting capacity of a given power system, without compromising on equity and fairness between different units.

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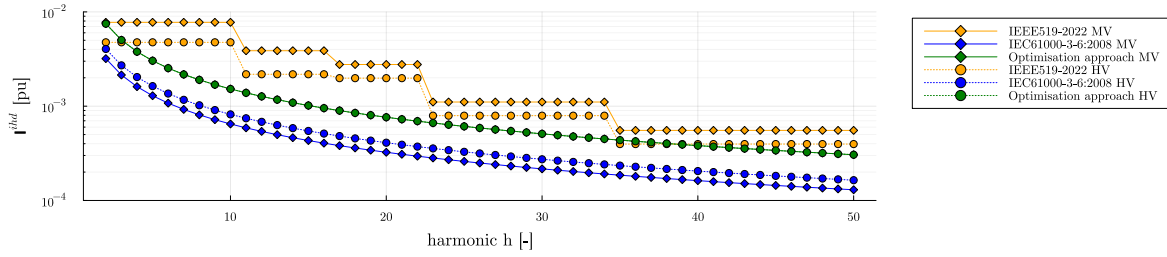


Fig. 2. Case A: inductive network – individual harmonic current $i_{12,h}^{ihd}$ of unit 12 at medium-voltage bus 17 (solid lines w. diamond marker) and $i_{3,h}^{ihd}$ unit 3 at high-voltage bus 4 (dotted lines w. circle marker), using one of three methods: a) IEEE519-2022 (orange), b) IEC61000-3-6:2008 (blue), and c) novel optimization method (green).

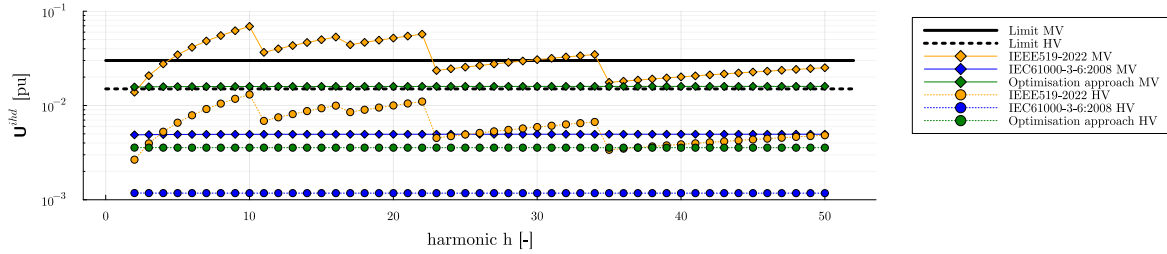


Fig. 3. Case A: inductive network – resulting individual harmonic voltage $U_{17,h}^{ihd}$ at medium-voltage bus 17 (solid lines w. diamond marker) and $U_{4,h}^{ihd}$ at high-voltage bus 4 (dashed lines w. circle marker), following a harmonic power flow based on the injected harmonic unit current determined using one of three methods: a) IEC61000-3-6:2008 (blue), b) IEEE519-2022 (orange), and c) novel optimization method (green).

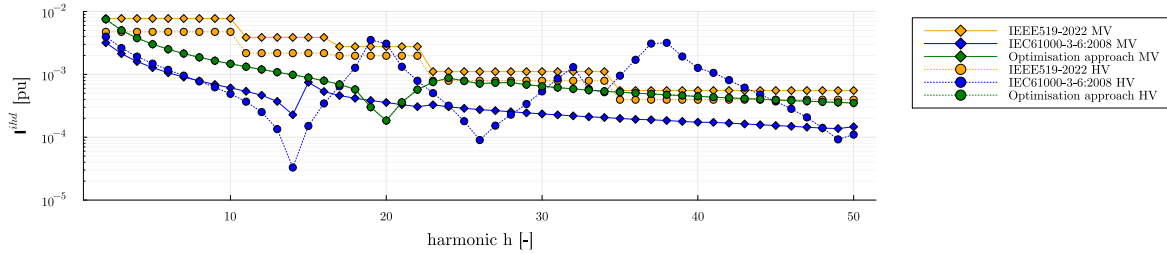


Fig. 4. Case B: underground cable network – individual harmonic current $i_{12,h}^{ihd}$ of unit 12 at medium-voltage bus 17 (solid lines w. diamond marker) and $i_{3,h}^{ihd}$ unit 3 at high-voltage bus 4 (dotted lines w. circle marker), using one of three methods: a) IEEE519-2022 (orange), b) IEC61000-3-6:2008 (blue), and c) novel optimization method (green).

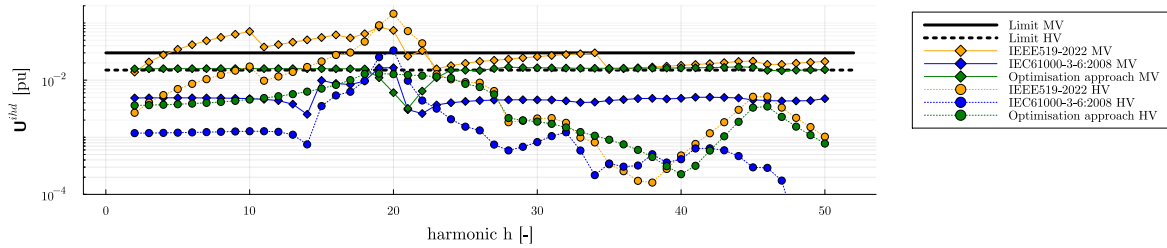


Fig. 5. Case B: underground cable network – resulting individual harmonic voltage $U_{17,h}^{ihd}$ at medium-voltage bus 17 (solid lines w. diamond marker) and $U_{4,h}^{ihd}$ at high-voltage bus 4 (dashed lines w. circle marker), following a harmonic power flow based on the injected harmonic unit current determined using one of three methods: a) IEC61000-3-6:2008 (blue), b) IEEE519-2022 (orange), and c) novel optimization method (green).